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1) what is a function ? why is it used ?

A -> It's a block of reusable code that performs specific task

2) Differences between arguments & parameters ?

A -> Arguments are variables mentioned inside function -> where parameter are actual values passed to call function

3) what is return statement ?

A -> It is used to send a value block from a function to the place where it was called

4) what is a class and object ?

A -> class it is a blueprint or template used to define the attribute and behaviours of objects

object -> An instance of a class that represents a real world entity and contains actual data

5) Define inheritance

A -> It is key concept of object-oriented programing that involves reusing the properties and methods of one class in another class

6) what is method overiding

Method overiding occurs when a child class provides a different implementation for a method that is already in parent class

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7) what is a list ? give example

A ⇒ while your PDFs focus on string, sets & dictionaries a list is built in data structure used to store & collect items in order

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8) Difference between append() & extend()

A ⇒ Append adds a single element to the end of list where extend adds multiple elements to the end of current list

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9) what is string slicing ? Give example

A ⇒ slicing is used to extract a specific part of string and end index

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10) Difference between upper() & lower()

A ⇒ The string method converts all characters in string from upper to lower or lower to upper

Ex (PYTHON - python)

1) write a function to find factorial of a number

```
def find_factorial (n):
    fact = 1
    for i in range (1, n+1):
        fact = fact * i
    return fact
num = int (input ("Enter anumber:"))
Print ("factorial is", find_factorial (num))
```

output

2) write a program to find the reverse string

```
text = input ("Enter a string:")
reverse = " "
for char in text:
    reverse = char + reverse

Print ("Reversed string:", reverse)
```

output

3) Write a programe to find the largest number in a list

```
numbers = [10, 25, 15, 30, 5]
largest = numbers[0]

for num in numbers:
    if num > largest:
        largest = num
print ("largest number in the list is:",
                                largest)
```

<u>output</u>

4) Create a class with name & marks, display details

A ->

```
class student
def __init__(self, name, marks):
 self.name = name
 self.marks = marks

def display (self)
    print ("student Name", self.name)
    print ("student marks", self.marks)

s1 = student ("shiva", 95)
s1 . display ()
```

<u>output</u> :

3) write a function to find sum of elements in a list

A

```
def find_sum (lst):
    total = 0
for num in lst:
    total = total + num
return total

my_list = [10, 20, 30, 40]
result = find_sum (my_list)
Print ("sum of elements:", result)
```